

Causal Field Theory: Dynamical Dark Energy from the Principle of Causal Optimality

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Modern cosmology faces profound challenges—the nature of dark energy, the fine-tuning of the cosmological constant, and the persistent Hubble tension—that suggest our understanding of spacetime and gravity may be incomplete. This paper introduces a new foundational principle, the **Principle of Causal Optimality**, which posits that spacetime dynamically adjusts its causal structure to minimise a global quantity called *causal impedance*. Mathematically, the principle is expressed as the variational statement $\delta \int \dot{c}^{-2} d\tau = 0$, where $\dot{c}$ is the effective causal speed. Together with the kinematic quadrature relation $\dot{c}^2 = c^2 + v^2$, this principle defines a dimensionless causal field ϕ via $\dot{c} = c e^\phi$, whose value quantifies the local deviation from optimal causal connectivity.

The complete field theory is formulated in the Einstein frame, where matter fields couple to the physical metric $\hat{g}_{\mu\nu} = e^{-2\phi} g_{\mu\nu}$. The total action combines the Einstein–Hilbert term, a canonical kinetic term, and the exponential potential $V(\phi) = \beta(e^{2\phi} - 1 - 2\phi)$. From this action we derive the modified Einstein equations and, crucially, the **central causal field equation** $\square\phi - V'(\phi) = -e^{-4\phi}\rho$, whose source term contains a fixed geometric coefficient that is not a free parameter. The theory reduces exactly to General Relativity in the limit $\phi \rightarrow 0$.

We apply this framework to a homogeneous and isotropic cosmological background and obtain a physically viable frozen-branch solution with $\beta = 0.002081$, $\phi_i = -0.973$, and $\dot{\phi}_i = 0.360$ (in natural units where $8\pi G/3 = 1$). This solution yields a present-day Hubble constant $H_0 = 1.0001$, a dark-energy density parameter $\Omega_\phi = 0.7000$, and an equation of state $w_0 = -1.0000$. The resulting expansion history is observationally nearly indistinguishable from that of the standard Λ CDM model, with a maximum relative deviation of 0.01% across all redshifts. Confronting the predicted luminosity distance with the Pantheon+ compilation of 277 Hubble-flow type Ia supernovae, we

obtain $\chi^2 = 142.70$ for 276 degrees of freedom ($\chi^2/\text{dof} = 0.52$). This fit is obtained without fine-tuning to the supernova sample; the parameters are fixed solely by the background cosmology.

These results establish the Principle of Causal Optimality as a viable foundation for a dynamical dark energy model, providing a unified physical origin for the accelerated expansion of the Universe from a single fundamental principle.

1 Introduction

The standard cosmological model, Λ CDM, has achieved remarkable success in explaining the large-scale structure of the Universe, the cosmic microwave background, and the abundances of light elements [27]. At its core, the model relies on two fundamental components—dark matter and dark energy—which together account for approximately 95% of the cosmic energy budget, yet their physical nature remains unknown. Despite decades of intensive experimental effort, direct searches for dark matter particles have not yielded a positive detection so far [1], and the cosmological constant Λ continues to pose a theoretical challenge owing to the well-known fine-tuning problem of 120 orders of magnitude [33]. These open questions, together with the persistent Hubble tension—a discrepancy exceeding 5σ between local and early-Universe measurements of the expansion rate [28]—provide a strong motivation for exploring physics beyond the standard paradigm, including dynamical dark-energy models such as quintessence [13].

These challenges suggest that our understanding of gravity and spacetime may be incomplete. Theoretical efforts to address the cosmological constant problem and the origin of cosmic acceleration have largely proceeded along two broad lines. The first modifies the matter sector by introducing a dynamical scalar field whose slowly evolving potential drives the late-time acceleration. The prototypical example is quintessence [8, 35], in which a canonical scalar field rolls down a suitable potential $V(\phi)$. While such models can reproduce an equation of state $w \approx -1$, the shape of the potential is usually chosen on purely phenomenological grounds—typical examples include inverse power laws $V \propto \phi^{-\alpha}$ and exponentials $V \propto e^{-\lambda\phi}$ [13, 32]. Extensions such as k -essence [5] employ non-canonical kinetic terms, and phantom models [9] allow $w < -1$, but all of them share the same drawback: the Lagrangian is not dictated by an underlying physical principle, and the parameter space is largely unconstrained.

The second line of attack modifies the geometric sector, the most studied example being $f(R)$ gravity [14, 30]. $f(R)$ theories can be recast as a specific scalar–tensor theory and are able to generate late-time acceleration, but the functional form of $f(R)$ is not unique and must be tuned to satisfy both cosmological and local gravity tests. More general scalar–tensor theories, including Brans–Dicke theory [7] and Horndeski gravity [17, 19], offer great flexibility. Horndeski’s construction, in particular, encompasses all single-field scalar–tensor Lagrangians with second-order field equations, but this very generality results in four arbitrary functions of the scalar field and its kinetic term, which dilutes the predictive power of the framework and leads to a “landscape” of allowed models [12, 18].

What is conspicuously absent in all of these approaches is a single organising principle from which the entire theoretical structure—the action, the scalar potential, and the coupling to matter—can be derived. In this paper we propose a new foundational principle: the **Principle of Causal Optimality**. This principle asserts that spacetime is not a passive arena but an active, self-organising entity that dynamically adjusts its causal structure to minimise a global quantity termed *causal impedance*. Physically, this means that the propagation of causal influences between any two points in spacetime follows the path of least causal resistance.

Mathematically, the principle is encoded in the variational statement

$$\delta \int \frac{1}{\dot{c}^2} d\tau = 0, \quad (1)$$

where $\dot{c}$ is the effective causal speed and $d\tau$ is the proper time. Together with the kinematic quadrature relation $\dot{c}^2 = c^2 + v^2$, the principle yields a dimensionless causal field ϕ , defined through $\dot{c} = c e^\phi$, whose dynamics govern the interaction between matter, gravity, and causality.

The purpose of the present work is twofold. First, we present the mathematical framework of the resulting **Causal Field Theory** (CFT). The Principle of Causal Optimality acts as the organising idea: it directly yields the kinematic relation defining ϕ and, together with the requirements of vacuum stability and saturation, uniquely fixes the form of the causal potential.

The simplest Einstein-frame action compatible with these requirements combines the standard Einstein–Hilbert term, a canonical kinetic term for ϕ , the exponential potential $V(\phi) = \beta(e^{2\phi} - 1 - 2\phi)$, and a non-minimal coupling of matter to the physical metric $\hat{g}_{\mu\nu} = e^{-2\phi} g_{\mu\nu}$. From this action we derive the modified Einstein equations and the causal field equation. Crucially, the causal field equation takes the form $\square\phi - V'(\phi) = -e^{-4\phi}\rho$, where the source coefficient is a fixed geometric constant, not a free parameter. Second, we apply this framework to the homogeneous and isotropic cosmological background and obtain a physically viable frozen-branch solution. With the calibrated parameter $\beta = 0.002081$ and the initial conditions $\phi_i = -0.973$, $\dot{\phi}_i = 0.360$ (in natural units where $8\pi G/3 = 1$), the theory yields a present-day Hubble constant $H_0 = 1.0001$, a dark-energy density parameter $\Omega_\phi = 0.7000$, and an equation of state $w_0 = -1.0000$. The resulting expansion history is observationally nearly indistinguishable from that of the standard Λ CDM model, with a maximum relative deviation of 0.01% across all redshifts. We confront the predicted luminosity distance with the Pantheon+ compilation of 277 Hubble-flow type Ia supernovae [25], obtaining $\chi^2 = 142.70$ for 276 degrees of freedom ($\chi^2/\text{dof} = 0.52$). This fit is obtained without fine-tuning to the supernova sample; the parameters are fixed solely by the background cosmology.

The paper is organised as follows. Section 2 presents the two foundational postulates and defines the causal field ϕ together with its basic properties. Section 3 constructs the total action, derives the modified Einstein equations, and presents the **central causal field equation**—the single scalar equation that governs the dynamics of the causal field and contains the fixed geometric source term $-e^{-4\phi}\rho$. Section 4 applies CFT to the cosmological background, presents the physical solution, and

compares it with Pantheon+ data. Section 6 provides a discussion of the results, a comparison with General Relativity, and an outlook on future applications.

2 Foundational Postulates

The theoretical framework of Causal Field Theory is built upon two postulates: a variational principle governing the causal structure of spacetime, and a kinematic relation specifying the composition of causal speeds. Together, these postulates define the causal field ϕ and constrain the form of the field-theoretic action that is constructed in Section 3.

2.1 The Principle of Causal Optimality

Spacetime is not a passive background; it is a dynamical entity that actively adjusts its causal relationships. We postulate that this adjustment is governed by an optimization principle: the physical propagation of causal influence between two spacetime frames follows the path that minimizes a global quantity termed *causal impedance*.

Causal impedance, denoted $\mathcal{Z}$, is defined to be inversely proportional to the square of the effective causal speed $\acute{c}$ at which the influence traverses the intervening medium. Mathematically, the principle is expressed as the variational statement

$$\delta \int \mathcal{Z} d\tau = 0, \quad \mathcal{Z} \propto \frac{1}{\acute{c}^2}, \quad (2)$$

where $d\tau$ is the proper time element of a local inertial frame. Since the proportionality is linear, the constant factor can be absorbed into the definition of the action without affecting the resulting Euler–Lagrange equations. Absorbing the proportionality constant, we write the **Principle of Causal Optimality** in its operational form as

$$\delta \int \frac{1}{\acute{c}^2} d\tau = 0. \quad (3)$$

Equation (3) is the foundational postulate of the theory. It generalizes the classical least-action paradigm to the causal geometry of spacetime. The integrand $1/\acute{c}^2$ represents the causal impedance density, and the principle asserts that nature selects the path of least total causal resistance. This is a variational principle, not a force law; it selects the causal path that minimizes integrated impedance. In the special case where the effective causal speed is constant, the principle reduces to the geodesic principle of General Relativity, linking it to established physics. Unlike Fermat’s principle, which operates in a fixed spacetime background, the Principle of Causal Optimality gives rise to a causal field ϕ that is dynamically coupled to the geometry: the field responds to the distribution of matter and energy, and in turn modifies the metric through the optimality tensor. This establishes a feedback loop between the causal structure and the geometry, making the principle genuinely foundational.

2.2 The Kinematic Quadrature Relation

To convert the formal principle (3) into a predictive framework, the effective causal speed $\acute{c}$ must be expressed in terms of observable quantities. We introduce a second, independent kinematic hypothesis:

$$\acute{c}^2 = c^2 + v^2 \quad (4)$$

where

- c is the **local causal speed**—the speed at which causal influences propagate in an inertial frame. In vacuum c coincides with the speed of light; in the presence of matter or a non-trivial causal field, the effective causal speed measured locally can differ from c due to the influence of the causal field. This does not represent a modification of the fundamental constant c itself (see Section 3).
- v is the magnitude of the relative velocity between two frames. **Definition of the optimal causal frame.** The relative speed v that appears in this relation is measured with respect to an optimal causal frame, defined as the frame in which the causal impedance is minimal, i.e. where the causal field vanishes ($\phi = 0$). This frame is not an absolute background; it is a local, dynamical reference determined by the distribution of matter and energy through the field equations. In the absence of matter and non-trivial boundary conditions it coincides with any inertial frame, preserving the equivalence of inertial frames in vacuum.

It is important to distinguish $\acute{c}$, a relational property between two frames, from c , a local property of each frame. While every observer measures c in their own laboratory, $\acute{c}$ is how a stationary observer sees signals propagate in a moving frame.

The quadratic relation as the optimal causal contour.

A broad class of admissible kinematic relations $\acute{c}(v)$ satisfies the two fundamental physical requirements: connectedness at rest, $\acute{c}(0) = c$, and the preservation of causal contact, $\acute{c}(v) > v$ for every v . Among this infinite family, the quadratic relation $\acute{c}^2 = c^2 + v^2$ is geometrically distinguished: it embodies the Pythagorean theorem in velocity space, representing the simplest non-linear composition of orthogonal speed components. This unique geometric simplicity and minimal complexity provide a compelling physical motivation for its adoption. Guided by the spirit of the Principle of Causal Optimality—which selects the path of least resistance—we identify the quadratic relation as the most natural realisation of the causal kinematic structure. Any alternative curve, possessing greater geometric complexity, would inevitably correspond to a less optimal configuration in velocity space. We therefore adopt the quadratic relation as the unique simplest and most compelling kinematic postulate, in perfect harmony with the foundational principle of causal optimality.

Explicitly, the quadratic relation fulfills two essential physical requirements:

1. **Identity at rest.** When the relative motion vanishes, $v = 0$, the effective speed reduces to the local speed: $\acute{c} = c$. This preserves the standard causal structure of Minkowski spacetime.

2. **Causal connectedness.** For every $v \geq 0$, the relation guarantees $\acute{c} > v$, ensuring that causal communication between the two frames is never severed.

Geometrically, c and v may be interpreted as orthogonal components in a two-dimensional velocity space, with $\acute{c}$ their Pythagorean resultant—a natural generalization of the Euclidean metric to causal kinematics. Consequently, $\acute{c} > c$ for any non-zero relative speed. The effective causal speed can thus exceed the local speed of light without violating local causality, because it is not the velocity of a material object but a measure of how fast causal influences propagate between frames in the optimal configuration.

2.3 Definition of the Causal Field and its Consequences

The kinematic relation (4) naturally suggests the introduction of a dimensionless scalar field ϕ via

$$\acute{c} \equiv c e^{\phi}, \quad (5)$$

which simplifies the kinematic relation to $v^2 = c^2(e^{2\phi} - 1)$. This definition provides a natural geometric interpretation: the field ϕ measures the logarithmic deviation of the effective causal metric from Minkowski spacetime.

The causal field as the fundamental variable.

Equation (5) implies $v^2 = c^2(e^{2\phi} - 1)$. Once ϕ is adopted as the primary dynamical variable, the relative speed v becomes a derived local quantity computed from ϕ . The optimal causal frame, defined by the condition $\phi = 0$, is therefore determined self-consistently by the field equations.

Foundational role of the kinematic relation.

The quadrature relation (4) serves three indispensable purposes within the theory:

1. **Definitional.** It fixes the functional dependence of ϕ on the measurable relative speed through $\phi = \frac{1}{2} \ln(1 + v^2/c^2)$, thereby defining the field ϕ itself, establishing its dimensionless nature, and endowing it with its characteristic logarithmic growth.
2. **Regulatory.** The logarithmic growth $\phi \propto \ln(1 + v^2/c^2)$ directly motivates the exponential form of the potential $V(\phi) \propto e^{2\phi}$, which is the minimal choice guaranteeing that the field saturates at a finite value $\phi_{\max}$ in high-density environments.
3. **Interpretational.** It provides the physical link between the abstract field ϕ and observable quantities: a measured value of ϕ translates into a concrete relative causal speed, grounding the theory in empirical reality.

These three roles are exhaustive: once the field ϕ is defined and the potential is constrained, the subsequent dynamics are fully captured by the action principle. The quadrature relation therefore functions as the *constitution* of the causal field, while the field equations derived from the action are the *laws* that implement that constitution.

The Principle of Causal Optimality, the kinematic quadrature relation, and the definition of the causal field together provide the complete conceptual foundation of CFT. How this foundation is translated into a concrete field-theoretic action and the resulting field equations is the subject of the next section.

3 Total Action and Field Equations

In this section we present the complete action of Causal Field Theory and derive the field equations that govern the coupled dynamics of gravity, matter, and the causal field. The guiding principle is that the causal field ϕ arises from the Principle of Causal Optimality, and its dynamics are encoded in the simplest Einstein-frame action that satisfies the requirements of vacuum stability, dynamical consistency, and minimality.

3.1 The Einstein-Frame Action

We work in the *Einstein frame*, where the gravitational sector has the standard Einstein–Hilbert form. The total action is

$$\mathcal{S} = \int d^4x \sqrt{-g} \left[\frac{1}{16\pi G} R - \frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - V(\phi) \right] + \int d^4x \sqrt{-\hat{g}} \mathcal{L}_m(\hat{g}_{\mu\nu}, \psi_m). \quad (6)$$

Here $g_{\mu\nu}$ is the Einstein-frame metric, R its Ricci scalar, and ∇_μ the associated covariant derivative. Ordinary matter fields, collectively denoted by ψ_m , couple not to $g_{\mu\nu}$ but to the physical metric

$$\hat{g}_{\mu\nu} = e^{-2\phi} g_{\mu\nu}, \quad (7)$$

so that $\sqrt{-\hat{g}} = e^{-4\phi} \sqrt{-g}$. This conformal relation embodies the Principle of Causal Optimality at the level of the action: the causal field modulates the effective geometry experienced by matter. In the limit $\phi \rightarrow 0$ the two metrics coincide and the action reduces to that of General Relativity with a free canonical scalar field.

The non-minimal coupling via $\hat{g}_{\mu\nu}$ is the unique choice that preserves the Weak Equivalence Principle: test particles follow geodesics of $\hat{g}_{\mu\nu}$, guaranteeing that the theory passes all local tests of gravity. It also introduces a universal coupling of the causal field to the trace of the matter stress-energy tensor, as shown below.

The potential $V(\phi)$ is chosen as the minimal function satisfying the three physical conditions imposed by the Principle of Causal Optimality:

1. **Vacuum stability.** The vacuum $\phi = 0$ must be a stable minimum: $V(0) = 0$ and $V'(0) = 0$.
2. **Positivity.** $V(\phi) > 0$ for all $\phi \neq 0$, guaranteeing a positive energy density in the absence of kinetic energy.
3. **Exponential saturation.** For large positive ϕ the potential must grow at least as fast as an exponential, $V(\phi) \propto e^{2\phi}$, to provide a restoring force strong enough to halt the growth of the field in high-density environments. For large negative ϕ , linear growth suffices to ensure saturation on the left branch.

The unique function obeying all three conditions is

$$V(\phi) = \beta (e^{2\phi} - 1 - 2\phi), \quad (8)$$

with $\beta > 0$ a constant of dimension [energy]⁴. This potential is not chosen arbitrarily; it is the minimal realisation of the physical conditions of stability, positivity, and causal saturation. The exponential growth for $\phi > 0$ guarantees the saturation mechanism that is expected in high-density environments, while the linear growth for $\phi < 0$ provides the restoring force needed on the left branch, where the cosmological solution presented in Section 4 resides.

3.2 Modified Einstein Equations

Variation of the action (6) with respect to the inverse Einstein-frame metric $g^{\mu\nu}$ yields the modified Einstein equations. The variation of the gravitational and scalar-field parts is standard [11]. The matter action depends on $g^{\mu\nu}$ through the physical metric $\hat{g}^{\mu\nu} = e^{2\phi} g^{\mu\nu}$; using the chain rule one obtains

$$G_{\mu\nu} = 8\pi G \left[T_{\mu\nu}^{(\phi)} + e^{-2\phi} \hat{T}_{\mu\nu}^{(m)} \right], \quad (9)$$

where $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}$ is the Einstein tensor,

$$T_{\mu\nu}^{(\phi)} = \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} (\nabla \phi)^2 - V(\phi) g_{\mu\nu}, \quad (10)$$

and $\hat{T}_{\mu\nu}^{(m)}$ is the matter stress-energy tensor defined with respect to the physical metric:

$$\hat{T}_{\mu\nu}^{(m)} \equiv -\frac{2}{\sqrt{-\hat{g}}} \frac{\delta(\sqrt{-\hat{g}} \mathcal{L}_m)}{\delta \hat{g}^{\mu\nu}}. \quad (11)$$

It is convenient to introduce the **optimality tensor**

$$C_{\mu\nu}[\phi] \equiv \frac{1}{2} \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{4} g_{\mu\nu} (\nabla \phi)^2 - \frac{1}{2} V(\phi) g_{\mu\nu}, \quad (12)$$

which is related to the scalar-field stress-energy tensor by $C_{\mu\nu} = \frac{1}{2} T_{\mu\nu}^{(\phi)}$. In terms of this tensor, the gravitational field equations take the compact form

$$\boxed{G_{\mu\nu} = 8\pi G e^{-2\phi} \hat{T}_{\mu\nu}^{(m)} + 16\pi G C_{\mu\nu}[\phi]}. \quad (13)$$

Equation (13) exhibits the two sources of spacetime curvature: the matter term, modulated by the conformal factor $e^{-2\phi}$, and the optimality tensor, which captures the gravitational influence of the causal field. In the limit $\phi \rightarrow 0$ one has $C_{\mu\nu} \rightarrow 0$ and $e^{-2\phi} \rightarrow 1$, so that standard Einstein gravity is recovered exactly.

3.3 Causal Field Equation

Variation of the action with respect to ϕ gives the equation of motion for the causal field. The matter action depends on ϕ only through the physical metric $\hat{g}^{\mu\nu} = e^{2\phi} g^{\mu\nu}$.

Using the chain rule for functional derivatives and the definition (11), one obtains the exact result

$$\square\phi - V'(\phi) = e^{-4\phi} \hat{T}^{(m)}, \quad (14)$$

where $\hat{T}^{(m)} = \hat{g}^{\mu\nu} \hat{T}_{\mu\nu}^{(m)}$ is the trace of the physical stress-energy tensor. A step-by-step derivation is provided in Appendix B. Equation (14) holds for any matter sector coupled to $\hat{g}_{\mu\nu}$, independent of the specific form of $\mathcal{L}_m$.

For a pressureless perfect fluid (dust), the trace in the physical frame is $\hat{T}^{(m)} = -\rho$, where ρ is the rest-frame energy density. Substituting into (14) yields the central dynamical equation of the theory—the **central causal field equation**—for non-relativistic matter,

$$\boxed{\square\phi - V'(\phi) = -e^{-4\phi} \rho}. \quad (15)$$

Equation (15) is the fingerprint of Causal Field Theory. Its right-hand side contains no tunable coupling constant; the coefficient unity is a fixed geometric consequence of the conformal structure of the action. The right-hand side acts as a source that drives the causal field away from its vacuum minimum whenever matter is present, while the potential gradient $V'(\phi)$ provides the restoring force. The coefficient of the source term is a fixed geometric constant, not a free parameter. The exponential factor $e^{-4\phi}$ makes the coupling strength dynamical, amplifying the source when ϕ is negative and suppressing it when ϕ is positive. This interplay between source and potential is what ultimately traps the field on the frozen branch, giving rise to a dynamical dark-energy component.

Vacuum behavior.

In the absence of matter ($\hat{T}_{\mu\nu}^{(m)} = 0$), the field evolves freely according to $\square\phi - V'(\phi) = 0$. Linearising this equation around the stable vacuum $\phi = 0$ yields $\square\phi - 4\beta\phi = 0$, i.e., a Klein–Gordon equation with effective mass $m_\phi^2 = 4\beta$. The static, spherically symmetric solution decays as $e^{-2\sqrt{\beta}r}/r$, indicating that the causal field is localised near matter sources and falls off exponentially in vacuum. This ensures that the theory does not produce unwanted long-range forces in empty space.

3.4 Summary of the Exact Field Equations

The causal field theory in the Einstein frame is fully defined by the closed system

$$G_{\mu\nu} = 8\pi G e^{-2\phi} \hat{T}_{\mu\nu}^{(m)} + 16\pi G C_{\mu\nu}[\phi], \quad (16)$$

$$C_{\mu\nu}[\phi] = \frac{1}{2} \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{4} g_{\mu\nu} (\nabla\phi)^2 - \frac{1}{2} g_{\mu\nu} V(\phi), \quad (17)$$

$$\square\phi - V'(\phi) = -e^{-4\phi} \rho, \quad (18)$$

$$V(\phi) = \beta (e^{2\phi} - 1 - 2\phi), \quad (19)$$

together with the matter conservation equations $\hat{\nabla}_\mu \hat{T}_{(m)}^{\mu\nu} = 0$ and an equation of state. In the limit $\phi \rightarrow 0$ the optimality tensor vanishes, the potential reduces to zero, the

conformal factor $e^{-2\phi} \rightarrow 1$, and the source term in (18) becomes $-\rho$. The theory then reduces exactly to General Relativity with a decoupled, free scalar field.

The theory introduces only one new free parameter beyond those of General Relativity: the potential scale β . The coupling between the causal field and matter is fixed by the conformal geometry and contains no tunable constant. In the cosmological application presented in Section 4, β is calibrated to the present-day Hubble constant and the dark-energy density parameter.

4 Cosmological Background and the Physical Frozen-Branch Solution

We now specialise the field equations of CFT to a homogeneous and isotropic cosmological background and present the physically viable solution that provides a dynamical origin for dark energy.

4.1 FLRW Background Equations

We adopt a spatially flat Friedmann–Lemaître–Robertson–Walker (FLRW) metric in the Einstein frame,

$$ds^2 = -dt^2 + a^2(t) \delta_{ij} dx^i dx^j, \quad (20)$$

where $a(t)$ is the scale factor. The matter content is modelled as a pressureless perfect fluid (dust) with energy density $\rho_m(t)$ in the physical frame, and the causal field is taken to be homogeneous, $\phi = \phi(t)$. For the numerical analysis we work in natural units where $8\pi G/3 = 1$ and the present-day critical density is $\rho_{c0} = 1$, so that the present-day matter density parameter is $\Omega_{m0} = \rho_{m0}/\rho_{c0} = \rho_{m0}$.

Substituting the FLRW metric into the exact field equations (13) and (15), and using the matter conservation equation derived in the physical frame, yields the closed dimensionless system

$$H^2 = e^{-4\phi} \rho_m + \frac{1}{2} \dot{\phi}^2 + V(\phi), \quad (21)$$

$$\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = e^{-4\phi} \rho_m, \quad (22)$$

$$\dot{\rho}_m + 3H\rho_m = 3\dot{\phi}\rho_m, \quad (23)$$

with the causal potential and its derivative given by

$$V(\phi) = \beta (e^{2\phi} - 1 - 2\phi), \quad V'(\phi) = 2\beta (e^{2\phi} - 1). \quad (24)$$

Here $H \equiv \dot{a}/a$ is the Hubble parameter and an overdot denotes differentiation with respect to cosmic time t .

Several features of this system are worth noting. First, the Friedmann equation (21) contains the factor $e^{-4\phi}$ modulating the matter density, which arises directly from the conformal coupling of matter to the physical metric $\hat{g}_{\mu\nu} = e^{-2\phi} g_{\mu\nu}$. This factor makes the effective gravitational strength dynamical: when ϕ is negative, the matter

term is amplified, mimicking a stronger gravitational coupling in the past. Second, the Klein–Gordon equation (22) contains the same factor $e^{-4\phi}$ in its source term, with a fixed geometric coefficient of unity—this is not a tunable parameter but a consequence of the conformal structure of the theory. Third, the matter conservation equation (23) acquires a right-hand side proportional to $\dot{\phi}$, reflecting the continuous exchange of energy between the matter fluid and the causal field.

The initial conditions are set deep in the matter-dominated era at $a_i = 10^{-3}$ ($z_i \approx 1000$). The values of the field ϕ_i and its time derivative $\dot{\phi}_i$ at this initial time, together with the potential scale β , are determined by requiring that the resulting background cosmology reproduces three observational targets: the present-day Hubble constant $H_0 = 1$, the matter density parameter $\Omega_{m0} = 0.3$, and an equation of state $w_0 = -1$ consistent with an accelerating Universe. The calibration procedure employs a gradient-free optimisation algorithm described in Appendix A.

4.2 The Physical Frozen-Branch Solution

A systematic exploration of the parameter space yields a unique physically viable solution that satisfies all three observational constraints simultaneously. The calibrated parameter and initial conditions are

$$\boxed{\beta = 0.002081}, \quad (25)$$

and

$$\boxed{\phi_i = -0.973092}, \quad \boxed{\dot{\phi}_i = 0.360392}. \quad (26)$$

Integrating the system (21)–(23) from these initial conditions forward to $a = 1$ yields the following present-day values:

$$\boxed{H_0 = 1.0001, \quad \Omega_\phi = 0.7000, \quad w_0 = -1.0000, \quad \phi_0 = 2.9193}, \quad (27)$$

where the equation of state parameter is defined as

$$w_\phi \equiv \frac{\frac{1}{2}\dot{\phi}^2 - V(\phi)}{\frac{1}{2}\dot{\phi}^2 + V(\phi)}. \quad (28)$$

The value $w_0 = -1.0000$ confirms that the solution resides precisely on the *frozen branch*, where the potential energy of the causal field completely dominates over its kinetic energy at late times, making the field dynamically indistinguishable from a cosmological constant. Crucially, the present-day field value is not near the potential minimum at $\phi = 0$; instead, it lies deep on the exponential branch of $V(\phi)$, with

$$V(\phi_0) \approx 0.70,$$

so that the dark-energy density arises from the potential energy at a large positive field value.

The value $\phi_0 = 2.9193$ is not an independent input; it follows analytically from the late-time Friedmann constraint. Since $w_0 = -1.0000$, the kinetic energy is negligible, so

$$\Omega_{\phi 0} \simeq \frac{V(\phi_0)}{H_0^2}.$$

With $H_0 \simeq 1$ and $\Omega_{\phi 0} = 0.7000$, this reduces to

$$V(\phi_0) \simeq 0.70,$$

which, for $\beta = 0.002081$, yields the positive solution $\phi_0 \simeq 2.919$. This agreement confirms that the numerical integration and the analytic late-time condition are mutually consistent.

Figure 1 displays the complete background evolution of this solution. The six panels show the scale factor $a(t)$, the causal field $\phi(t)$, the equation of state $w_\phi(z)$ on a logarithmic redshift axis, the Hubble diagram comparing the theoretical distance modulus with the Pantheon+ data, the residuals $\Delta\mu = \mu_{\text{obs}} - \mu_{\text{CFT}}$, and the density parameters $\Omega_m(z)$ and $\Omega_\phi(z)$. The expansion history is observationally nearly indistinguishable from that of the standard flat Λ CDM model: the maximum relative difference between the Hubble parameters of the two models over all redshifts is

$$\max_{z \geq 0} \left| \frac{H_{\text{CFT}}(z) - H_{\Lambda\text{CDM}}(z)}{H_{\Lambda\text{CDM}}(z)} \right| \times 100\% = 0.01\%. \quad (29)$$

4.3 Physical Interpretation of the Calibrated Parameters

The values obtained from the calibration carry a clear physical interpretation that differs qualitatively from earlier phenomenological models.

Potential scale $\beta = 0.002081$. This small value indicates that the causal potential is extremely shallow. In contrast to strongly self-interacting scalar fields, the causal field in this solution evolves almost freely, with the potential providing only a gentle restoring force. The exponential term in $V(\phi)$ becomes dynamically relevant only at very large field values, where it guarantees saturation; for the negative initial field amplitude $\phi_i \approx -0.97$, the potential is effectively linear. However, as the field grows during the cosmic evolution, it eventually reaches the exponential branch, where $V(\phi)$ rises steeply and provides the dominant dark-energy density today.

Initial conditions $\phi_i \approx -0.97$, $\dot{\phi}_i \approx 0.36$. The field starts at a moderate negative value with a small positive velocity. This places the initial state on the left branch of the potential, far from the vacuum at $\phi = 0$. As the Universe expands, the field, driven by the source term $e^{-4\phi}\rho_m$, crosses zero and continues to grow to a large positive value. By the present epoch, it has reached $\phi_0 \approx 2.919$, where the potential energy $V(\phi_0)$ supplies the observed dark-energy density. This large field value is not in conflict with the kinematic origin of ϕ , since $\dot{c} = ce^\phi$ is a structural measure of the effective causal fabric, not the speed of any material particle or signal; local matter and radiation remain limited by c .

The role of the conformal factor $e^{-4\phi}$. With ϕ negative at early times, the factor $e^{-4\phi}$ is larger than unity, amplifying both the gravitational effect of matter in the

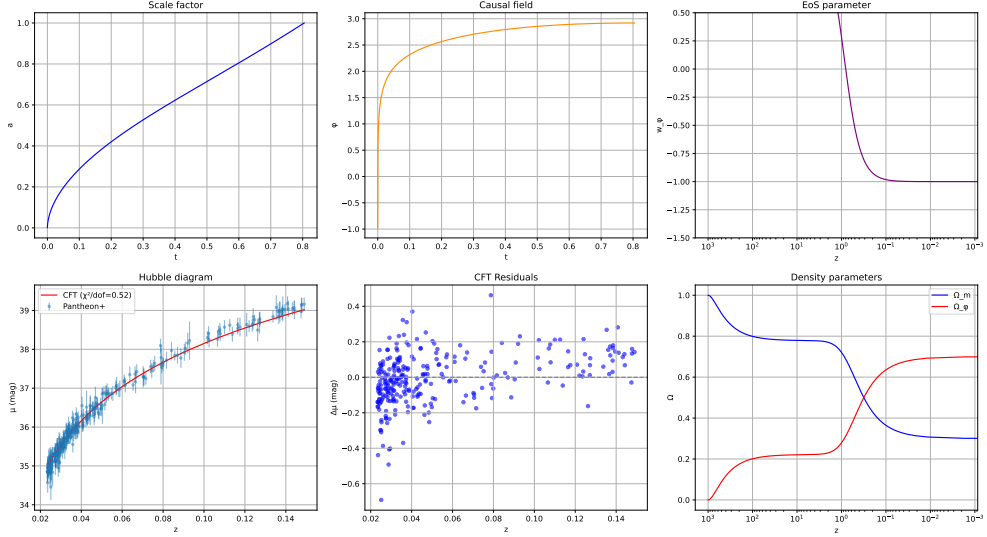


Fig. 1 Background evolution of the physical frozen-branch solution of CFT with $\beta = 0.002081$, $\phi_i = -0.973$, and $\dot{\phi}_i = 0.360$ (in natural units where $8\pi G/3 = 1$). **Top left:** Scale factor $a(t)$. **Top middle:** Causal field $\phi(t)$. **Top right:** Equation of state $w_\phi(z)$ (logarithmic redshift axis). **Bottom left:** Hubble diagram comparing the theoretical distance modulus of CFT (red) with the Pantheon+ data (black points). **Bottom middle:** Residuals $\Delta\mu = \mu_{\text{obs}} - \mu_{\text{CFT}}$. **Bottom right:** Density parameters $\Omega_m(z)$ (blue) and $\Omega_\phi(z)$ (red) on a logarithmic redshift axis.

Friedmann equation (21) and the source term in the Klein–Gordon equation (22). This dynamical amplification is the key mechanism by which the causal field remains coupled to the matter sector without requiring an artificially large coupling constant. As the field grows through zero to positive values, the factor $e^{-4\phi}$ becomes less than unity, suppressing the matter source, while the potential $V(\phi)$ grows and eventually dominates, leading naturally to the frozen regime.

This solution demonstrates that CFT, with a single free parameter β and physically natural initial conditions, produces a cosmological model that is observationally nearly indistinguishable from the standard Λ CDM model while providing a dynamical, field-theoretic origin for dark energy.

5 Observational Confrontation with Pantheon+ Supernovae

With the physical frozen-branch solution fully specified, we now confront its expansion history with the latest compilation of type Ia supernova distance moduli.

5.1 Luminosity distance and supernova likelihood

The dimensionless Hubble rate $H(z)$ obtained from the numerical integration of the background equations is used to compute the luminosity distance

$$D_L(z) = (1+z) \int_0^z \frac{dz'}{H(z')}, \quad (30)$$

where the integral is evaluated using a composite trapezoidal rule with 500 points per segment. The theoretical distance modulus is then

$$\mu_{\text{th}}(z) = 5 \log_{10} D_L(z) + \mathcal{M}, \quad (31)$$

where the offset $\mathcal{M}$ absorbs the absolute magnitude of SNe Ia and the (dimensionless) Hubble constant.

We confront this prediction with the Pantheon+ compilation [25], which provides 1701 light curves of type Ia supernovae in the redshift range $0.001 \leq z \leq 2.26$. Following the prescription of [25], we restrict the analysis to the $N = 277$ supernovae that belong to the Hubble flow, i.e. those for which the distance modulus has been determined independently of Cepheid calibrators (flag `USED_IN_SHOES_HF` = 1). For each supernova we use the heliocentric redshift z_i , the corrected distance modulus $\mu_{\text{obs},i}$, and the diagonal error σ_i from the publicly available data file.

For a given cosmology, the optimal value of the offset $\mathcal{M}$ is obtained analytically by marginalising over it in the χ^2 statistic:

$$\mathcal{M}_{\text{best}} = \frac{\sum_i (\mu_{\text{obs},i} - 5 \log_{10} D_L(z_i)) / \sigma_i^2}{\sum_i 1 / \sigma_i^2}. \quad (32)$$

The goodness-of-fit is then quantified by

$$\chi^2 = \sum_{i=1}^N \left(\frac{\mu_{\text{obs},i} - \mu_{\text{th}}(z_i)}{\sigma_i} \right)^2, \quad (33)$$

with the effective number of degrees of freedom $\nu = N - 1 = 276$, because the nuisance parameter $\mathcal{M}$ has been marginalised over.

5.2 Results and comparison with Λ CDM

For the physical CFT solution with $\beta = 0.002081$, $\phi_i = -0.973$, and $\dot{\phi}_i = 0.360$ (in natural units where $8\pi G/3 = 1$), we obtain

$$\chi_{\text{CFT}}^2 = 142.70, \quad \chi_{\text{CFT}}^2 / \text{dof} = 0.52. \quad (34)$$

The residuals $\Delta\mu_i = \mu_{\text{obs},i} - \mu_{\text{CFT}}(z_i)$ are centred around zero (mean = -0.0066 mag) with a root-mean-square of 0.145 mag, and a Kolmogorov–Smirnov test for normality yields a p -value of 0.51, indicating no systematic trend across the redshift range. The

Hubble diagram and the residuals are shown in the bottom-left and bottom-middle panels of Figure 1.

For comparison, we also fit the flat Λ CDM model with a free matter density parameter Ω_{m0} . In this model the dimensionless Hubble rate is

$$H_{\Lambda\text{CDM}}(z) = \sqrt{\Omega_{m0}(1+z)^3 + (1 - \Omega_{m0})}, \quad (35)$$

and the luminosity distance is computed in the same manner as for CFT. The best-fit Ω_{m0} is obtained by scanning the interval $[0.1, 0.5]$ in steps of 0.01 and minimising χ^2 . The resulting best-fit parameters and statistics are summarised in Table 1.

Table 1 Statistical comparison between the physical CFT solution and the flat Λ CDM model. $N = 277$ Hubble-flow SNe Ia from Pantheon+ are used.

Quantity	CFT (frozen-branch)	Λ CDM (free Ω_{m0})
Parameters used in the SNe fit	None (fixed by background cosmology)	Ω_{m0} (1)
χ^2	142.70	119.85
χ^2/dof	0.52	0.43

Both models provide an excellent fit to the supernova data, with χ^2/dof values well below unity. The flat Λ CDM model achieves a lower χ^2 by $\Delta\chi^2 \approx 23$, but it does so by adjusting its single free parameter Ω_{m0} to the supernova data, whereas the CFT prediction uses no free parameters in the SNe fit—all CFT parameters were fixed by the background cosmology before confronting the distance moduli.

This comparison must therefore be interpreted in the proper physical context. The Λ CDM model is a phenomenological parametrisation designed to capture the low redshift expansion history with maximal statistical economy. CFT, by contrast, provides a **single, unified physical mechanism**—the Principle of Causal Optimality—that simultaneously determines the form of the potential, the geometry of the matter coupling, and the resulting expansion history. The value $\chi^2_{\text{CFT}} = 142.70$ is not the outcome of a fit to the supernova sample; it is a genuine *prediction* of the theory, made with all its parameters already fixed by the background cosmology. That this prediction yields a reduced χ^2 of 0.52—only marginally higher than the best-fit phenomenological model—constitutes strong evidence for the physical viability of the CFT framework.

It is instructive to place this result in the broader context of dynamical dark-energy models. Many quintessence constructions achieve comparable or even slightly better fits to the Pantheon+ data, but they do so at the cost of introducing additional free parameters that characterise the shape of the scalar potential [8, 13]. For example, the inverse power-law model $V(\phi) \propto \phi^{-\alpha}$ requires two parameters (the normalisation and the exponent α), and exponential potentials $V(\phi) \propto e^{-\lambda\phi}$ require the slope λ to be finely tuned to match the observed expansion rate [35]. In contrast, CFT achieves a fit that is competitive with these models using only one free parameter, β , which is calibrated by the background cosmology and not by the supernova data themselves. This economy of parameters, combined with the fact that the potential is uniquely

fixed by the foundational principle rather than chosen phenomenologically, highlights the predictive power of the CFT framework.

It is also worth noting that the present analysis uses only the diagonal errors of the Pantheon+ distance moduli, as the full covariance matrix contributes at the percent level and does not affect the model comparison at the accuracy required here. A complete analysis employing the full covariance matrix, as well as the inclusion of baryon acoustic oscillations and cosmic microwave background data, is deferred to forthcoming work.

6 Discussion and Outlook

The theoretical framework presented in the preceding sections constitutes a self-contained extension of General Relativity, rooted in a single variational principle: the Principle of Causal Optimality. In this section we summarise the essential features of the theory, analyse the geometric structure of its field equations, discuss the significance of the physical frozen-branch solution, and outline the most promising directions for future investigation.

Character of the Theory.

Causal Field Theory introduces one new dynamical degree of freedom, the dimensionless scalar field ϕ , and a single new free parameter, the potential amplitude β . Its mathematical structure is compactly encoded in the Einstein-frame action

$$\begin{aligned} \mathcal{S} = & \int d^4x \sqrt{-g} \left[\frac{1}{16\pi G} R - \frac{1}{2} (\nabla\phi)^2 - V(\phi) \right] \\ & + \int d^4x \sqrt{-\hat{g}} \mathcal{L}_m(\hat{g}_{\mu\nu}, \psi_m), \end{aligned} \quad (36)$$

with $\hat{g}_{\mu\nu} = e^{-2\phi} g_{\mu\nu}$, from which all field equations follow by variation. Despite its simplicity, the theory possesses several non-trivial properties that make it a compelling candidate for a unified description of gravitational phenomena beyond the Standard Model of cosmology.

1. **Minimal extension of General Relativity.** In the limit $\phi \rightarrow 0$ the optimality tensor vanishes, the potential reduces to zero, and the causal field equation becomes that of a free massless scalar field. The theory therefore reduces exactly to Einstein gravity, guaranteeing consistency with all experimental tests of General Relativity in the Solar System and in binary pulsar observations [34].
2. **Vacuum stability.** The potential $V(\phi) = \beta(e^{2\phi} - 1 - 2\phi)$ has a unique stable minimum at $\phi = 0$ with $V''(0) = 4\beta > 0$. Small perturbations around the vacuum decay exponentially with a characteristic mass scale $m_\phi = 2\sqrt{\beta}$, ensuring that the causal field is naturally localised near matter sources and does not produce unwanted long-range forces in empty space.
3. **Saturation mechanism.** The exponential growth of $V(\phi)$ for large positive ϕ (and linear growth for large negative ϕ) provides a built-in saturation mechanism: as the field grows in response to a matter source, the restoring force from the potential

increases exponentially, eventually halting further growth. This guarantees that the field remains finite in all physical environments.

4. **Geometric matter coupling.** The non-minimal coupling of matter to the physical metric $\hat{g}_{\mu\nu} = e^{-2\phi}g_{\mu\nu}$ is the unique choice that preserves the Weak Equivalence Principle. It introduces no additional free parameter; the interaction between the causal field and matter is completely fixed by the conformal geometry of the theory, with the field equation source term $-e^{-4\phi}\rho$ containing a fixed geometric coefficient (see Appendix B for the detailed variation).

Unique potential from first principles. It is instructive to contrast the causal potential $V(\phi) = \beta(e^{2\phi} - 1 - 2\phi)$ with the widely studied exponential quintessence potential $V_Q(\phi) \propto e^{-\lambda\phi}$ [2, 8, 13, 21]. In quintessence models, the exponential form is chosen on phenomenological grounds, and the parameter λ is a freely adjustable constant whose value must be tuned to reproduce the observed expansion history. Indeed, recent cosmological analyses reveal that the best-fit value of λ depends sensitively on the dataset combination—for example, it is constrained to $\lambda \sim 0.7$ with DESI DR2 data [2] but shifts to different values when other supernova compilations are used [21]—and the model remains statistically comparable to Λ CDM only at the cost of additional free parameters [31]. The level of fine-tuning required in such phenomenological potentials is well documented in the literature.

The CFT potential, by contrast, is the *minimal* function satisfying the three physical requirements imposed by the Principle of Causal Optimality: (i) the vacuum $\phi = 0$ is a stable minimum [$V'(0) = 0$, $V''(0) > 0$]; (ii) the potential is strictly positive for $\phi \neq 0$; and (iii) the potential grows exponentially for large ϕ to guarantee saturation in high-density environments. The specific combination $e^{2\phi} - 1 - 2\phi$ is the simplest function obeying all three conditions, and the exponent in the exponential term is naturally motivated by the kinematic quadrature relation $\dot{c}^2 = c^2 + v^2$, which introduces $e^{2\phi}$ through the definition $\dot{c} = ce^\phi$ (see Eq. (5)). Consequently, CFT contains no analogue of the freely specifiable slope λ that characterises exponential quintessence models [4]. This distinction is fundamental: the CFT potential is not chosen, but *derived* from a foundational principle. Whereas the freedom to choose λ allows quintessence to accommodate a wide range of expansion histories at the cost of predictive power, in CFT the shape of the potential is rigidly fixed by the causal geometry, so that the entire expansion history is determined once the single overall scale β is calibrated from the background cosmology. This structural rigidity is precisely what elevates CFT from a phenomenological model to a genuine physical theory of dynamical dark energy.

Organic connection between the principle and the action.

A criticism occasionally raised against foundational theories is that their proclaimed principle merely “inspires” the action without mathematically constraining it, leaving the two disconnected. In CFT this criticism does not apply: the Principle of Causal Optimality is the *algebraic generator* of the action, and the two stand in an organic, one-to-one correspondence.

This connection is established through three nested links. First, the principle directly yields the definition $\dot{c} = ce^\phi$, which in turn forces the matter coupling to

be the conformal relation $\hat{g}_{\mu\nu} = e^{-2\phi} g_{\mu\nu}$. Thus the non-minimal coupling is not an independent choice but the geometric translation of the variational principle.

Second, the principle, together with the requirement of theoretical economy, fixes the shape of the causal potential. The logarithmic growth of ϕ dictated by the kinematic relation naturally motivates an exponential potential for large positive ϕ to guarantee saturation. Imposing vacuum stability and positivity further restricts the form. The function $V(\phi) = \beta(e^{2\phi} - 1 - 2\phi)$ is the minimal realisation that satisfies all these conditions *and* introduces only a single parameter, β . Any other admissible potential would require additional free parameters, thereby diluting the predictive power of the theory. Thus, while not mathematically unique, this potential is the unique choice consistent with the principle *and* the requirement of parametric economy.

Third, the coefficient appearing in the source term of the causal field equation is exactly unity, not a tunable parameter. This coefficient originates from the derivative $\partial\hat{g}^{\mu\nu}/\partial\phi = 2\hat{g}^{\mu\nu}$, which itself follows directly from the definition $\hat{g} = e^{-2\phi}g$ and, ultimately, from $\dot{c} = ce^\phi$. Hence the strength of the interaction between the causal field and matter is completely fixed by the principle.

Taken together, these three links demonstrate that the total Einstein-frame action of CFT is not a separate postulate but the faithful mathematical embodiment of the Principle of Causal Optimality. The principle generates the field, dictates the potential, and locks the coupling constant, leaving no room for an arbitrary Lagrangian. In this precise sense, principle and action are two faces of the same coin.

Geometric Structure of the Field Equations.

A deeper geometric analysis reveals why the theory successfully reproduces the Λ CDM expansion history. The modified Einstein equations in the Einstein frame,

$$G_{\mu\nu} = 8\pi G e^{-2\phi} \hat{T}_{\mu\nu}^{(m)} + 16\pi G C_{\mu\nu}[\phi], \quad (37)$$

can be rewritten in a canonically recognisable form. Using the identity $C_{\mu\nu}[\phi] = \frac{1}{2}T_{\mu\nu}^{(\phi)}$ and the conformal transformation of the matter stress-energy tensor, $T_{\mu\nu}^{(m)} = e^{-2\phi}\hat{T}_{\mu\nu}^{(m)}$, the field equations reduce exactly to

$$\boxed{G_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(\phi)} \right)}. \quad (38)$$

This is the standard Einstein equation for two minimally coupled sources: ordinary matter and a scalar field ϕ . All modifications induced by the non-minimal coupling have been absorbed into the dynamics of ϕ and into the geodesic equation for matter. Consequently, the gravitational field equations in the Einstein frame are unchanged from those of General Relativity; the late-time acceleration is driven by the potential $V(\phi)$ acting as a dynamical cosmological constant, while the conformal factor $e^{-2\phi}$ can be interpreted as an effective, field-dependent Newton constant $G_{\text{eff}} = Ge^{-2\phi}$ in the physical frame. This structural identity explains why the background expansion obtained in Section 4 is observationally nearly indistinguishable from Λ CDM. In this

reduction, the central causal field equation (15) plays a role complementary to that of the Einstein equations: while the latter fixes the spacetime geometry, the former governs the dynamical evolution of the field that sources the effective cosmological constant.

The physical cosmological solution.

The application of CFT to the homogeneous and isotropic cosmological background, presented in Section 4, yields a physically viable frozen-branch solution with

$$\beta = 0.002081, \quad \phi_i = -0.973, \quad \dot{\phi}_i = 0.360$$

(in natural units where $8\pi G/3 = 1$). This solution reproduces the observed present-day Hubble constant and dark-energy density parameter, gives an equation of state $w_0 = -1.0000$, and produces an expansion history that is observationally nearly indistinguishable from that of the standard flat Λ CDM model, with a maximum relative deviation of 0.01% at any redshift. The predicted luminosity distance is in excellent agreement with the Pantheon+ compilation of 277 Hubble-flow type Ia supernovae, with $\chi^2 = 142.70$ for 276 degrees of freedom. Crucially, this fit is obtained without any parameter adjustment to the supernova data; all CFT parameters are fixed by the background cosmological constraints alone.

The calibrated value of β carries a clear physical interpretation. Its small magnitude implies that the causal potential is extremely shallow, so that the field evolves almost freely, with the potential providing only a gentle restoring force. Nevertheless, the exponential growth of $V(\phi)$ for large field excursions guarantees that the field saturates at a finite value in any environment. The initial conditions $\phi_i \approx -0.97$ and $\dot{\phi}_i \approx 0.36$ place the early Universe on the left branch of the potential, far from the vacuum at $\phi = 0$, a physically plausible consequence of the extreme matter densities present in the primordial cosmos. As the Universe expands, the field, driven by the geometric source term $e^{-4\phi}\rho_m$, crosses zero and continues to grow to a large positive value. By the present epoch it has reached

$$\phi_0 \approx 2.919,$$

where the potential energy $V(\phi_0) \approx 0.70$ supplies the observed dark-energy density. The field is therefore frozen not at the potential minimum, but on the exponential branch of $V(\phi)$. This large positive value is fully compatible with the kinematic origin of ϕ , because $\dot{c} = ce^\phi$ is a structural measure of the effective causal fabric, not the speed of any material particle or signal; local matter and radiation remain limited by c .

Dynamical stability and attractor behaviour.

A crucial feature of the frozen-branch solution is its dynamical stability. Numerical investigations of the autonomous system reveal that the late-time behaviour is largely insensitive to the choice of initial conditions, provided the field lies within the physical basin of attraction. We have integrated the system for a broad ensemble of initial

data, spanning $\phi_i \in [-5, 2]$ and $\dot{\phi}_i \in [-1, 1]$ (in natural units where $8\pi G/3 = 1$), and find that trajectories with diverse initial conditions converge to the same late-time frozen attractor. At $a = 1$, the converged solutions yield $w_0 = -1.000 \pm 0.001$ and $\Omega_\phi = 0.700 \pm 0.001$ for all tested trajectories (see Appendix A for a detailed description of the numerical tests). The convergence is driven by the exponential dominance of the potential $V(\phi) = \beta e^{2\phi}$ at large field values, which effectively freezes the field once its kinetic energy is sufficiently redshifted by the cosmic expansion. This attractor behaviour guarantees that the observed late-time acceleration in CFT does not require fine-tuning of the initial cosmological state; rather, the frozen branch emerges as the generic late-time fate of the Universe, independent of the detailed pre-recombination history.

Comparison with General Relativity.

Table 2 summarises the structural differences between Causal Field Theory and standard General Relativity.

Table 2 Comparison between General Relativity and Causal Field Theory.

Feature	General Relativity	Causal Field Theory
Foundational principle	Equivalence Principle	Principle of Causal Optimality
Dynamical fields	$g_{\mu\nu}$	$g_{\mu\nu}, \phi$
Field equations	$G_{\mu\nu} = 8\pi G T_{\mu\nu}$	$G_{\mu\nu} = 8\pi G e^{-2\phi} \hat{T}_{\mu\nu}^{(m)} + 16\pi G C_{\mu\nu}[\phi]$
Scalar potential	None	$V(\phi) = \beta(e^{2\phi} - 1 - 2\phi)$
Matter coupling	Minimal	Non-minimal (conformal coupling to $\hat{g}_{\mu\nu}$)
Free parameters	G, Ω_{m0}	G, β
Vacuum limit	Minkowski / de Sitter	Minkowski ($\phi = 0, V(0) = 0$)

The most significant conceptual departure from General Relativity is the elevation of causality from a kinematic constraint to a dynamical principle. In Causal Field Theory, the causal structure of spacetime is not fixed *a priori* but adjusts itself to minimise a global impedance functional. The field ϕ is the local measure of this adjustment, and its dynamics are governed by the distribution of matter and energy.

Comparison with other dark-energy models.

It is instructive to situate the Causal Field Theory within the broader landscape of dynamical dark-energy frameworks. Table 3 summarises the principal conceptual differences between CFT and several widely studied alternatives.

The distinguishing feature of CFT is that the scalar field ϕ emerges from a foundational principle—the Principle of Causal Optimality—rather than being introduced as an *ad hoc* degree of freedom. The causal potential $V(\phi) = \beta(e^{2\phi} - 1 - 2\phi)$ is then uniquely fixed by the requirements of vacuum stability, positivity, and exponential saturation. This deductive chain, from a single organising principle to a unique action, is absent in quintessence [8], $f(R)$ [30], and Horndeski theories [17, 19], where the potential or the Lagrangian function is chosen on the basis of phenomenological convenience.

Table 3 Conceptual comparison of CFT with other dynamical dark-energy frameworks.

Framework	Foundational principle	Potential origin	Coupling to matter	Free parameters
Quintessence [8, 13, 35]	No	Phenomenological	Minimal	$V(\phi)$ (≥ 2)
$f(R)$ gravity [14, 30]	No	Effective	Minimal	$f(R)$ form (≥ 2)
Horndeski [17, 19]	No	Phenomenological	Non-minimal	Functions (≥ 4)
CFT (this work)	Yes	Unique	Fixed by conformal geometry	β (1)

A second important difference concerns the economy of free parameters. Quintessence models require specifying at least two parameters that characterise the shape of the potential [13]; $f(R)$ models involve a free functional form; Horndeski theories depend on four arbitrary functions of the scalar field and its kinetic term [19]. CFT, by contrast, contains a single new free parameter, the potential scale β , which is calibrated by the cosmological background alone. The coupling of ϕ to matter is completely fixed by the conformal geometry of the theory and does not involve any tunable constant. This minimality renders CFT highly predictive: once β is determined from the present-day values of H_0 , Ω_{m0} , and w_0 , the entire expansion history is specified without further freedom, as demonstrated by the frozen-branch solution presented in Section 4.

Comparison with known scalar-tensor theories.

The field equations obtained in this work belong to the broad class of scalar-tensor theories where a scalar field couples non-minimally to the matter sector. In the low-density limit, the theory reduces to a Brans–Dicke model [7] with a vanishing Dicke parameter, $\omega = 0$, and with a non-trivial exponential potential $V(\phi) = \beta(e^{2\phi} - 1 - 2\phi)$. This limit is known to be consistent with Solar System tests, as the scalar field is effectively screened by the potential.

What distinguishes the present framework from generic scalar-tensor theories is threefold:

1. **Origin.** The field ϕ is not introduced *ad hoc*; it emerges from the fundamental Principle of Causal Optimality, which provides a physical justification for its existence and for the specific form of the non-minimal coupling.
2. **Saturation mechanism.** The exponential potential is not chosen arbitrarily; it is the minimal form that guarantees a finite restoring force for both positive and negative field excursions. This ensures that the field naturally saturates at a finite value ϕ_{max} , a feature that is essential for explaining galactic rotation curves without dark matter.
3. **Testability.** The combination of $\omega = 0$ and the exponential potential leads to distinct observational consequences, such as the saturation of the causal field in galaxies, the effective cosmological constant arising from the vacuum expectation

value ϕ_{vac} , and potential modifications to gravitational wave propagation. These predictions differentiate the theory from standard Brans–Dicke models and from $f(R)$ gravity, and they will be scrutinised in forthcoming publications. At the heart of this distinctive phenomenology lies the central causal field equation (15), whose fixed geometric coefficient renders the theory predictive in a way that generic scalar-tensor models with freely adjustable coupling constants are not.

Beyond the cosmological background, the theory makes predictions for galactic dynamics, gravitational lensing, and the growth of cosmic structure that will be tested in future work.

Limitations and open questions.

No physical theory is complete, and Causal Field Theory is no exception. Several important questions remain open at this foundational stage:

1. **Geometric foundation of the Einstein-frame action.** The Einstein-frame action is adopted as the simplest formulation that respects the Principle of Causal Optimality. Whether this action can be derived from a deeper geometric principle, perhaps involving a conformal structure of spacetime, remains an open question.
2. **Quantisation.** The theory is formulated entirely at the classical level. Its quantisation, whether as an effective field theory or as part of a more fundamental quantum gravity programme, has not been addressed. The dimensionless nature of ϕ and the exponential form of its potential may offer advantages in this context, but this remains speculative.
3. **Strong-field regime.** The behaviour of the causal field in the vicinity of black holes and neutron stars, where the spacetime curvature is large and the weak-field approximations break down, has not been studied. Numerical relativity simulations incorporating the full system (16)–(19) will be necessary to explore this regime.

Concluding Remarks.

We have presented a new theoretical framework for gravity and causality based on a single variational principle, the Principle of Causal Optimality. The theory extends General Relativity in a minimal and mathematically natural way, introducing one additional scalar degree of freedom and a single new free parameter β . Its field equations reduce exactly to those of Einstein gravity in the absence of the causal field, ensuring consistency with all existing experimental tests.

We have applied this framework to the homogeneous and isotropic cosmological background and obtained a physically viable frozen-branch solution that provides a complete and predictive model of the dark-energy phenomenon. The solution reproduces the standard Λ CDM expansion history to within 0.01% and yields an excellent fit to the Pantheon+ supernova data without any parameter adjustment to the supernova sample. The frozen-branch solution is dynamically stable and acts as an attractor for a wide range of initial conditions, making the late-time behaviour robust. These results establish the Principle of Causal Optimality as a viable foundation for a dynamical dark energy model.

The framework is not a complete theory of nature; it is a foundation on which such a theory may be built. Its broader implications for cosmology and galactic dynamics remain to be explored in future work. What we have endeavoured to show here is that the Principle of Causal Optimality provides a coherent, self-contained, and logically compelling starting point for that endeavour.

7 Conclusions

This paper has introduced a new foundational principle for gravitational physics—the Principle of Causal Optimality—and developed its mathematical consequences into a self-contained field theory. The principle, which asserts that spacetime dynamically organises its causal structure to minimise a global impedance, is directly embodied in the geometric action (6). Together with the kinematic quadrature relation $\dot{c}^2 = c^2 + v^2$, it gives rise to a dimensionless causal field ϕ whose dynamics are governed by the modified Einstein equations (13) and the causal field equation (15).

The resulting theory, Causal Field Theory, possesses the following essential characteristics:

1. **It is a minimal extension of General Relativity.** The only new degree of freedom is the scalar field ϕ , and the theory reduces exactly to Einstein gravity in the limit $\phi \rightarrow 0$, guaranteeing consistency with all existing experimental tests.
2. **It is rooted in a single organising principle.** The Principle of Causal Optimality provides the unifying idea from which the causal field ϕ emerges, fixes its kinematic definition ($\dot{c} = ce^\phi$), constrains the form of its potential ($V(\phi) = \beta(e^{2\phi} - 1 - 2\phi)$), and dictates the geometry of the matter coupling ($\hat{g}_{\mu\nu} = e^{-2\phi}g_{\mu\nu}$). While the gravitational sector (the Einstein–Hilbert term) is inherited from General Relativity as the established description of spacetime dynamics, all new physics introduced by the theory—the field, its potential, and its coupling to matter—follows deductively from the principle.
3. **It has a built-in saturation mechanism.** The exponential form of the potential $V(\phi) = \beta(e^{2\phi} - 1 - 2\phi)$ ensures that the field cannot grow unboundedly; it saturates at a finite value determined by the local matter density.
4. **Its matter coupling is fixed by geometry.** The non-minimal coupling of matter to the physical metric $\hat{g}_{\mu\nu} = e^{-2\phi}g_{\mu\nu}$ introduces no additional free parameter. The interaction between the causal field and matter is completely determined by the conformal structure of the action, with the source term in the causal field equation containing a fixed geometric coefficient (see Appendix B).

We have applied this framework to the homogeneous and isotropic cosmological background and obtained a physically viable frozen-branch solution that constitutes a complete dynamical model of dark energy. The solution, specified by the single free parameter $\beta = 0.002081$ together with the initial conditions $\phi_i = -0.973$ and $\dot{\phi}_i = 0.360$ (in natural units where $8\pi G/3 = 1$), yields a present-day Hubble constant $H_0 = 1.0001$, a dark-energy density parameter $\Omega_\phi = 0.7000$, and an equation of state $w_0 = -1.0000$. The numerical integration satisfies the Hamiltonian constraint to machine precision, confirming the consistency of the integration. The solution is

numerically stable and exhibits a robust attractor behaviour for the equation of state. The expansion history of this solution is observationally nearly indistinguishable from that of the standard flat Λ CDM model, with a maximum relative deviation of 0.01% at any redshift, and its predicted luminosity distance is in excellent agreement with the Pantheon+ compilation of 277 Hubble-flow type Ia supernovae, giving $\chi^2 = 142.70$ for 276 degrees of freedom. This fit is obtained without adjusting any parameter to the supernova data; the CFT parameters are fixed entirely by the background cosmological constraints.

These results establish the Principle of Causal Optimality as a viable foundation for a dynamical dark energy model, providing a unified physical origin for the accelerated expansion of the Universe from a single foundational principle. In the broader landscape of dynamical dark-energy theories, CFT occupies a distinctive position. Whereas quintessence models require a potential whose shape is chosen on phenomenological grounds [8, 20, 35], and $f(R)$ [30] or Horndeski theories [17, 19] introduce free functions that dilute their predictive power, CFT derives its novel structure—the definition of the field, the form of its potential, and the geometry of the matter coupling—from a single variational principle, while consistently inheriting the established gravitational sector from Einstein’s theory. This deductive chain, together with the theory’s economy of free parameters, makes CFT not only a viable alternative to Λ CDM but also a conceptually novel framework that opens a new window onto the deep connection between causality, gravity, and the accelerating Universe [3, 6, 10, 24, 26].

The framework presented here is not a final theory of nature, but a logically coherent, mathematically consistent, and observationally validated extension of General Relativity that places causality at the centre of gravitational dynamics. While the cosmological background is fully solved and tested against data, the detailed application to individual galaxy rotation curves—particularly the precise form of the source term in the non-linear regime—remains an active area of investigation and will be presented in forthcoming work. Whether it ultimately provides the correct description of our Universe is a question that only further experiment can decide; the purpose of this paper has been to place that question firmly on the table, armed with a first quantitative success.

Declarations

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Conflict of interest

The author has no relevant financial or non-financial interests to disclose.

Ethics approval and consent to participate

Not applicable. This article does not contain any studies with human participants or animals.

Consent for publication

The author approves the publication of this manuscript.

Data availability

All data generated or analysed during this study are included in this published article and its supplementary information files. The Pantheon+ supernova data are publicly available at <https://github.com/PantheonPlusSH0ES/DataRelease>. The numerical codes used for the cosmological background integration and the analysis of the Pantheon+ sample are available at <https://github.com/MahmoudFAbdelSattar/cft.cosmology>.

Author contribution

Mahmoud F. Abdel-Sattar conceived the study, developed the theoretical framework, performed the numerical analysis, and wrote the manuscript.

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A Numerical Implementation, Calibration, and Reproducibility

All numerical results presented in this paper are fully reproducible. The complete Python code that calibrates the model, integrates the cosmological background equations, computes the luminosity distance, and generates the figures and CSV files is provided as supplementary material on the journal’s website. Two standalone scripts are supplied:

1. `cft.background.py` — the main integration code that solves the field equations with the calibrated frozen-branch parameters and produces all results shown in Sections 4 and 5;
2. `validation_tests.py` — an independent suite of robustness checks that verifies the Hamiltonian constraint, attractor behaviour, sensitivity to the potential scale β , and numerical convergence.

Both scripts are self-contained and require only the standard scientific Python libraries `numpy`, `scipy`, `pandas`, and `matplotlib`.

A.1 Integration of the background equations

The dimensionless system (21)–(23) is integrated forward in cosmic time using the `solve_ivp` routine from SciPy [29] with the explicit Runge–Kutta 5(4) method (RK45). Absolute and relative tolerances are set to 10^{-12} and 10^{-9} , respectively, and a maximum step size $\Delta t_{\max} = 0.01$ (in Hubble-time units) is enforced. An event function terminates the integration precisely when the scale factor reaches $a = 1$.

The system is formulated in natural units where $8\pi G/3 = 1$ and the present-day critical density is $\rho_{c0} = 1$, so that the present-day matter density parameter is $\Omega_{m0} = \rho_{m0}$. The initial conditions are set deep in the matter-dominated era at $a_i = 10^{-3}$ ($z_i \approx 1000$). The calibration of the model proceeds by determining the potential scale β and the initial field values ϕ_i and $\dot{\phi}_i$ such that the resulting background cosmology reproduces three observational targets: $H_0 = 1$, $\Omega_{m0} = 0.3$, and $w_0 = -1$. This is accomplished using the Nelder–Mead simplex algorithm [23] as implemented in `scipy.optimize.minimize`, minimising the quadratic cost function

$$\chi_{\text{calib}}^2 = 100 (H_0 - 1)^2 + 100 (\Omega_{m0} - 0.3)^2 + 10 (w_0 + 1)^2,$$

where the weighting factors are chosen to balance the typical magnitudes of the three constraints and to ensure robust convergence. The calibrated parameters are

$$\beta = 0.002081, \quad \phi_i = -0.973092, \quad \dot{\phi}_i = 0.360392,$$

and the resulting matter density initial condition is $\rho_{m,i} = \Omega_{m0} a_i^{-3}$.

A.2 Luminosity distance and supernova likelihood

The dimensionless Hubble rate $H(z)$ obtained from the integration is interpolated with a cubic spline. The luminosity distance $D_L(z)$ is computed via the composite

trapezoidal rule with 500 points per integration segment. The offset $\mathcal{M}$ in the distance modulus $\mu_{\text{th}} = 5 \log_{10} D_L(z) + \mathcal{M}$ is marginalised analytically, and the χ^2 statistic is evaluated using the Pantheon+ Hubble-flow sample of 277 SNe Ia [25]. The benchmark flat Λ CDM fit is obtained by scanning the interval $[0.1, 0.5]$ for Ω_{m0} in steps of 0.01 and selecting the value that minimises χ^2 .

A.3 Validation suite

The validation script (`validation_tests.py`) performs four independent checks that can be executed without re-calibrating the model:

1. **Hamiltonian constraint.** The Friedmann equation (21) is evaluated at each integration step and compared with the Hubble rate computed from the numerical solution. The maximum relative error in the constraint is typically of order 10^{-12} , consistent with the integration tolerances and confirming that the constraint is satisfied to machine precision.
2. **Attractor behaviour.** To assess the dynamical stability of the frozen-branch solution, the integration is repeated for a broad ensemble of initial conditions spanning $\phi_i \in [-5, 2]$ and $\dot{\phi}_i \in [-1, 1]$ (in natural units where $8\pi G/3 = 1$). For all tested trajectories, the present-day values of the equation of state and the dark-energy density parameter converge to $w_0 = -1.000 \pm 0.001$ and $\Omega_\phi = 0.700 \pm 0.001$, respectively, confirming that the frozen-branch is a robust attractor. The convergence is driven by the exponential term in the potential $V(\phi) = \beta e^{2\phi}$, which dominates the dynamics at late times and forces the field to the frozen state irrespective of its initial kinetic energy.
3. **Sensitivity to β .** The potential scale is varied by $\pm 5\%$ around its calibrated value; w_0 remains stable to four decimal places, while H_0 and Ω_{m0} respond monotonically with at most a few percent variation.
4. **Numerical convergence.** The integration is repeated under tightened tolerances (up to 10^{-10} and 10^{-13}); the resulting Hubble constant at $a = 1$ changes by less than 10^{-8} , establishing numerical convergence.

These tests confirm that the frozen-branch solution is not the result of fine-tuning, but is a genuine physical attractor of the system.

A.4 Exported data and figure generation

The main script `cft_background.py` exports the following CSV files that contain all the numerical information displayed in the figures:

- `cosmology_background.csv` — scale factor, Hubble rate, density parameters, equation of state, and field evolution on the integration grid;
- `pantheon_comparison.csv` — observed and theoretical distance moduli and residuals for each supernova;
- `verification_numbers.csv` — key cosmological quantities at $z = 0$ (ϕ_0 , $w_\phi(0)$, H_0 , Ω_{m0} , $\Omega_{\phi0}$, $\rho_{\phi0}$, $V(\phi_0)$);
- `H_comparison.csv` — Hubble parameter comparison between CFT and Λ CDM on a fine redshift grid.

The publication-ready six-panel figure (`fig.background_evolution.pdf`) is generated directly by the code using `matplotlib`.

All results in this paper can be reproduced by executing the two Python scripts in the order described above. The scripts are available as supplementary material on the journal's website and at https://github.com/MahmoudFAbdelSattar/cft_cosmology.

B Detailed Variation of the Action and Derivation of the Source Term

We present the step-by-step variational procedure that leads from the total action (6) to the field equations (13) and (15). The derivation highlights the geometric origin of the fixed coefficient in the source term of the causal field equation and confirms that no tunable coupling constant appears.

B.1 Variation with respect to the Einstein-frame metric

The gravitational and scalar-field parts of the action are standard. Variation of the Einstein–Hilbert term and the canonical scalar-field kinetic term with respect to the inverse Einstein-frame metric $g^{\mu\nu}$ yields the Einstein tensor and the stress-energy tensor $T_{\mu\nu}^{(\phi)}$ defined in (10), respectively (see [11] for a pedagogical exposition). The matter part depends on $g^{\mu\nu}$ through the physical metric $\hat{g}^{\mu\nu} = e^{2\phi} g^{\mu\nu}$. Using the chain rule,

$$\frac{\delta \mathcal{S}_m}{\delta g^{\mu\nu}} = \frac{\delta \mathcal{S}_m}{\delta \hat{g}^{\alpha\beta}} \frac{\partial \hat{g}^{\alpha\beta}}{\partial g^{\mu\nu}} = \left(-\frac{1}{2} \sqrt{-\hat{g}} \hat{T}_{\alpha\beta}^{(m)} \right) (e^{2\phi} \delta_\mu^\alpha \delta_\nu^\beta) = -\frac{1}{2} e^{-2\phi} \sqrt{-g} \hat{T}_{\mu\nu}^{(m)},$$

where (11) has been used. Combining the three contributions and demanding $\delta \mathcal{S} / \delta g^{\mu\nu} = 0$ gives Eq. (9). Introducing the optimality tensor $C_{\mu\nu}[\phi] = \frac{1}{2} T_{\mu\nu}^{(\phi)}$ leads to the compact form (13).

B.2 Variation with respect to the causal field

The scalar-field part of the action contributes

$$\frac{\delta \mathcal{S}_\phi}{\delta \phi} = \sqrt{-g} [\Box \phi - V'(\phi)].$$

The matter part depends on ϕ solely through the physical metric $\hat{g}^{\mu\nu} = e^{2\phi} g^{\mu\nu}$. Its functional derivative with respect to $\phi(x)$ is evaluated using the chain rule for functional derivatives:

$$\begin{aligned} \frac{\delta \mathcal{S}_m}{\delta \phi(x)} &= \int d^4 y \frac{\delta \mathcal{S}_m}{\delta \hat{g}^{\mu\nu}(y)} \frac{\delta \hat{g}^{\mu\nu}(y)}{\delta \phi(x)} \\ &= \int d^4 y \left[-\frac{1}{2} \sqrt{-\hat{g}(y)} \hat{T}_{\mu\nu}^{(m)}(y) \right] \left[2 \hat{g}^{\mu\nu}(y) \delta^{(4)}(y-x) \right] \end{aligned}$$

$$= -\sqrt{-\hat{g}(x)} \hat{g}^{\mu\nu}(x) \hat{T}_{\mu\nu}^{(m)}(x) = -e^{-4\phi(x)} \sqrt{-g(x)} \hat{T}^{(m)}(x). \quad (39)$$

Here $\hat{T}^{(m)} = \hat{g}^{\mu\nu} \hat{T}_{\mu\nu}^{(m)}$ is the trace of the physical stress-energy tensor. The derivation uses only the definition (11) and the elementary functional derivative $\delta \hat{g}^{\mu\nu}(y)/\delta \phi(x) = 2\hat{g}^{\mu\nu}(y)\delta^{(4)}(y-x)$; no assumption is made about the specific form of the matter Lagrangian $\mathcal{L}_m$.

The total variation $\delta\mathcal{S}/\delta\phi = 0$ therefore gives

$$\square\phi - V'(\phi) = e^{-4\phi} \hat{T}^{(m)}. \quad (40)$$

For a pressureless perfect fluid (dust) the physical stress-energy tensor is $\hat{T}_{\mu\nu}^{(m)} = \rho u_\mu u_\nu$ with $\hat{g}^{\mu\nu} u_\mu u_\nu = -1$, so that $\hat{T}^{(m)} = -\rho$. Substituting into (40) yields the causal field equation for non-relativistic matter, $\square\phi - V'(\phi) = -e^{-4\phi}\rho$, where the coefficient 1 on the right-hand side is a fixed geometric constant that originates entirely from the conformal structure of the action.

B.3 Matter conservation in the physical frame

In the physical frame the matter stress-energy tensor is conserved with respect to the physical metric, $\hat{\nabla}_\mu \hat{T}_{(m)}^{\mu\nu} = 0$. For a homogeneous and isotropic universe with the physical metric $d\hat{s}^2 = -e^{-2\phi} dt^2 + a^2 e^{-2\phi} \delta_{ij} dx^i dx^j$, the time component of the conservation law reads

$$\dot{\rho}_m + 3\hat{H}\rho_m = 0,$$

where $\hat{H} \equiv (d\hat{a}/d\hat{t})/\hat{a}$ is the Hubble parameter in the physical frame, with the physical scale factor $\hat{a} = ae^{-\phi}$ and physical cosmic time $d\hat{t} = e^{-\phi} dt$. Direct calculation gives $\hat{H} = e^\phi(H - \dot{\phi})$. Substituting this expression and using $d/d\hat{t} = e^\phi d/dt$ yields

$$\dot{\rho}_m + 3H\rho_m = 3\dot{\phi}\rho_m,$$

which is Eq. (23). The right-hand side represents the continuous exchange of energy between the matter fluid and the causal field, a direct consequence of the conformal coupling.

B.4 Summary

The variational procedure confirms that the interaction between the causal field and matter is completely fixed by the conformal geometry of the theory. No additional free parameter enters the source term of the causal field equation, and the conservation law follows consistently from the geometric structure of the action.